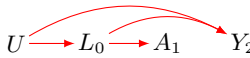
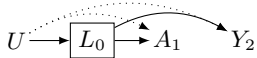
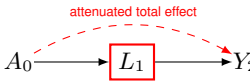
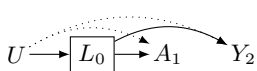
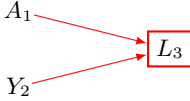
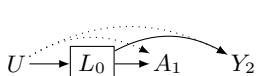
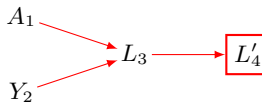
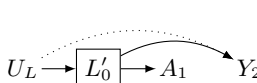


Five Elementary Applications of Pearl's Back-Door Adjustment Theorem

Rule	Problem	Solution
1	Ensure Causal Order: Timing of $A_{\phi 1}$ and $Y_{\phi 2}$ are incorrectly asserted; in truth, Y_1 induces an association with A_2. 	
2	Block back-door path by conditioning on a common cause or its proxy: A and Y share both measured and unmeasured common causes; conditioning to block the open backdoor path. 	
3	Do not condition on a mediator: $\boxed{L}$ blocks the total causal effect of $A \rightarrow Y$. If L may be affected by A, ensure L occurs before A. 	
4	Do not condition on a collider: $\boxed{L}$ induces a non-causal association between A and Y. Ensure L occurs before A and that A occurs before Y. 	
5	Conditioning on a descendant is akin to conditioning on its parent: $\boxed{L'}$, a descendant of an unmeasured confounder U, may also be affected by A. Ensure L' occurs before A. If L' is not affected by A or Y, L''s timing relative to A and Y is unimportant. We must only ensure that U occurs before A and that A occurs before Y. 	

Key:

A denotes the treatment; Y denotes the outcome; U denotes an unmeasured confounder; L denotes a confounder; $\rightarrow$ asserts causality; t subscript denotes the true relative timing of the variable; ϕt relative timing is **asserted**, here erroneously; $\rightarrow$ indicates a path for bias linking A to Y absent causation; $- \rightarrow$ indicates a path for bias separating A and Y ; $\boxed{L}$ indicates that conditioning on L introduces bias (over-conditioning bias). We include $A \leftarrow - U - \rightarrow Y$ to clarify that we cannot typically be confident that all common causes of the treatment and outcome have been measured. Examples 1, 3-5 illustrate how bias arises from erroneous variable timing: $\phi t \neq t$